

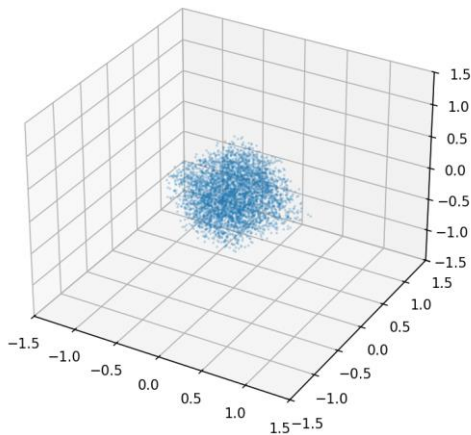
AP1

First I make 2 random 4-vectors in range(0,100) and i make 2x2 matrixes out of it with simple math operations.

```
for a in range(4):  
    i=int(a/2)  
    j=int(a%2)  
    matrix1[i][j]=vector1[a]  
    matrix2[i][j]=vector2[a]
```

Than I wrote functions for norms. I chose 1 and infinity norms for both vectors and matrixes and they are induced. Furthermore, I wrote two functions for calculating distances between matrixes and vectors. At the and I wrote function to draw ball which has norm as a parameter. You give which norm ball you want and it draws ball with center (0,0). We have 4-vectors but we just ignore 4-th element and than we can plot them on 3D.

Norm1:



Norm_inf:

